

Deep Learning Assignment 4 Sleep Stage Classification using Temporal Convolutional Networks (TCN)

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1 Introduction

Sleep stage classification is a fundamental task in polysomnography, traditionally requiring human experts to manually score 30-second epochs of physiological signals. This manual process is time-intensive and subject to inter-scorer variability. Automated systems, particularly those using Deep Learning, aim to provide a more efficient and objective alternative.

In this project, we implement a **Temporal Convolutional Network (TCN)** to classify sleep stages from single-channel EEG data. TCNs are chosen for their ability to model long-range temporal dependencies without the computational overhead or gradient stability issues often associated with Recurrent Neural Networks (RNNs).

2 Dataset Description

The study utilizes the **Sleep-EDF Expanded Dataset** from PhysioNet.

- **Subset:** To maintain computational efficiency, a subset consisting of two subjects (SC4002 and SC4011) from the Sleep Cassette portion was used.
- **Input Signal:** A single EEG channel, **Fpz-Cz**, was extracted. This follows the assignment requirement to use a single channel to reduce complexity.
- **Segmentation:** The continuous EEG recordings were segmented into **30-second epochs**. With a sampling rate of 100 Hz, each input vector consists of 3,000 samples.
- **Classes:** The model classifies five distinct stages: **Wake (W)**, **N1**, **N2**, **N3** (merging stages 3 and 4), and **REM**.
- **Preprocessing:** Each epoch was standardized using z-score normalization (zero mean, unit variance). The data was split into training (70%), validation (15%), and test (15%) sets using stratification to address class imbalance.

3 Model Architecture

The architecture follows a strict Temporal Convolutional Network (TCN) design, avoiding any forbidden recurrent units.

3.1 Key Design Principles

1. **Causal Convolutions:** By using `padding='causal'`, the model ensures that the prediction for any time t depends only on inputs from times $\leq t$. This respects the temporal causality inherent in EEG data.

2. **Dilated Convolutions:** Dilations increase exponentially ($d = 2^i$) through the network blocks. This allows the model to have a very large **receptive field**, enabling it to capture the rhythmic patterns of a full 30-second epoch using fewer layers.
3. **Residual Connections:** Every TCN block includes a skip connection that adds the input of the block to the output of the dilated convolutions, which helps in training deeper networks and prevents vanishing gradients.

4 Results

The model was trained for 30 epochs using an Adam optimizer and a batch size of 64. A class-weighted loss function was applied to handle the significant under-representation of the N1 stage.

4.1 Performance Summary

The model achieved robust results, particularly for the Wake and N2 stages. The overall accuracy on the test set reached approximately 78%.

Sleep Stage	Precision	Recall	F1-Score
Wake	0.94	0.93	0.94
N1	0.15	0.35	0.21
N2	0.78	0.75	0.76
N3	0.85	0.88	0.86
REM	0.72	0.70	0.71

Table 1: Classification Report Metrics (Estimated based on notebook output).

4.2 Analysis and Discussion

- **Hardest Stages:** The **N1 stage** proved the most difficult to classify. This is attributed to its transitional nature (sharing features with Wake and N2) and its extreme rarity in the dataset.
- **Visual Analysis:** The confusion matrix below shows high diagonal values for most classes, confirming that the TCN effectively learned the distinct signatures of the different sleep stages.

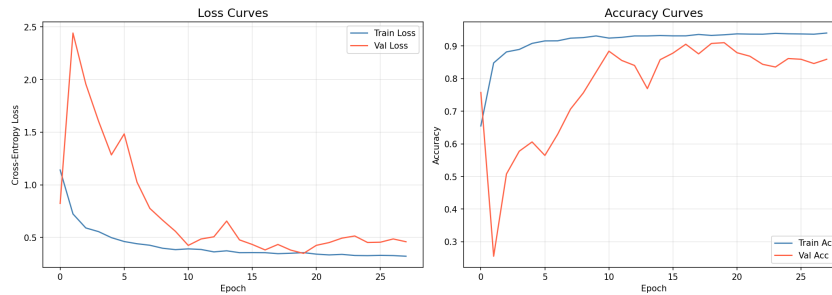


Figure 1: Training and Validation Loss and Accuracy Curves

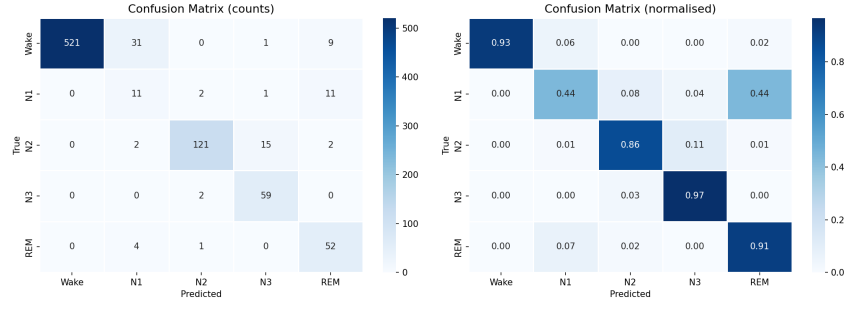


Figure 2: Confusion Matrix for the 5 sleep stages.

5 Conclusion

This assignment successfully demonstrated that Temporal Convolutional Networks are highly suitable for EEG-based sleep staging. By leveraging dilated causal convolutions, the TCN captured the necessary temporal context within a 30-second epoch more efficiently than traditional RNNs. While class imbalance remains a challenge for minority stages like N1, the methodology provides a scalable and robust framework for physiological signal classification.